

Shared mineral pathology in malignant and non-malignant lung tissues of non-smoking women from Xuan Wei, China

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ABSTRACT

Household coal smoke was evidenced to cause the uniquely high lung cancer mortalities in Xuan Wei, China. Lung cancer screening in local villagers found concurrently high prevalence of interstitial pneumonia. What could be the shared hazard linking those pathologies? Could it be some coal-derived mineral deposit in lungs that causes both fibrotic and malignant changes? A case-control study was conducted by recruiting 50 lung cancer cases and 30 pseudotumor controls among non-smoking women from Xuan Wei, China. The severity of interstitial lung abnormalities (ILAs) was compared between cases and controls by high-resolution computed tomography (HRCT) and histopathological analysis; mineral deposits in lung tissues were examined by light microscopy and electron microscopy techniques. The lung cancer case group and the pseudotumor group share this characteristic of abundant ILAs while the fibrotic changes are more severe in lung cancer than pseudotumor cases. Fibrotic changes are more severe in peribronchial lymph nodes than in lung parenchyma. Mineral deposits embedded in the anthracotic pigments are visible under polarized light microscopy (PLM) due to their birefringence. The morphology of this birefringent mineral resembles that of acicular berthierine-chamosite identified in Late Permian coal from Xuan Wei, China. These preliminary findings underscore the need for further investigation into the mineralogical factors contributing to the high incidence of lung cancer in this region.

1. Introduction

Xuan Wei, a rural county situated in Yunnan Province of southwest China, has received extensive research attention for the highest lung cancer mortality (especially among females who are often non-smoking housewives) for decades (Li et al., 2019; Lin et al., 2015; Mumford et al., 1987). Indoor air pollution from the burning of “smoky” coal was established as the main cause based on sufficient evidence in both humans and experimental animals (Lan et al., 2002; Liang et al., 1988).

Polycyclic aromatic hydrocarbons (PAHs) formed from the incomplete combustion of bituminous (smoky) coal have been previously studied as the key toxic constituents contributing to the elevated lung cancer risk in Xuan Wei (Mumford et al., 1987, 1995; Vermeulen et al., 2019). Yet, significant reduction in indoor PAHs, achieved through interventions such as chimney installation, did not effectively curb lung cancer mortality in the region, which continued to rise at a higher pace than in other rural areas of China (Ren et al., 2016; Tian et al., 2009). We suspect that some other etiological agent(s) could have been overlooked.

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Along with lung cancer, interstitial pneumonia appears to be prevalent in Xuan Wei. Responding to villagers' complaints to newspapers on the extremely high rates of lung cancer, the government of Yunnan Province conducted lung cancer screening by chest X-ray in a village named Hutou in 2005. Among 169 villagers examined, 10 (5.9 %) incident lung cancer and 15 (8.9 %) interstitial pneumonia cases were diagnosed (Yunnan Provincial Health Commission., 2005). The concurrently elevated detection rates of interstitial pneumonia and lung cancer in this village suggests some fibrosis-inducing coal smoke component as the potentially shared etiological factor underlying these two pathologies.

One such candidate could be an asbestos-like mineral in Late Permian coal from Xuan Wei, China. In Xuan Wei, the highest lung cancer rates are concentrated in areas that produce and utilize the uppermost Late Permian C1 coal (Dai et al., 2008). The distinctive physicochemical and mineralogical feature of C1 coal is its high concentration of fine-grained quartz and the iron-rich silicate minerals (Dai et al., 2008; Large et al., 2009). Recent findings from geochemists indicate that the iron-rich mineral from those Late Permian coals of Xuan Wei turned out to be interstratified berthierine-chamosite (Kogure et al., 2019; Li et al., 2021). Berthierine belongs to the same serpentine mineral group as chrysotile asbestos which can increase the risks of lung scarring (inflammation and fibroplasia) and lung cancer (Bouros et al., 2002). In comparison with chrysotile, berthierine can be biologically more active because it has an octahedral sheet containing essentially Fe^{2+} and not Mg^{2+} (Costa et al., 1990, 1989). This is echoed by one preliminary observation of excellent correlation between iron content in coal and lung cancer risks in different communes of Xuan Wei (Tian et al., 2002).

We hereby report a case-control study of lung cancer cases and pseudotumor controls among non-smoking women in Xuan Wei, comparatively characterizing the mineral deposits in sampled tissues of lung parenchyma and peribronchial lymph nodes, as well as examining the severity of interstitial lung abnormalities (ILAs) among participants. Featured by ground glass or reticular abnormalities, traction bronchiectasis, honeycombing, or non-emphysematous cysts through radiologic scans, ILAs can be indicative of subclinical or early pulmonary fibrosis and are associated with adverse disease progression (Chae et al., 2021; Hatabu et al., 2020). Our findings hold implications for offering insights into the etiology and pathogenesis of lung cancer in Xuan Wei, China, and informing early management and targeted control strategies.

2. Materials and methods

2.1. Case-control study design

This is a hospital-based case-control study to compare two distinct pathologies: lung cancer and pseudotumor of Xuan Wei, China. We refer to the lung cancer group as "cases" and the pseudotumor group as "controls". By comparing ILA severity and mineral deposits in lung tissues from both groups, we tested the hypothesis that the two pathologies of pseudotumor and lung cancer in this setting of extremely high lung cancer rates share exposure to one same kind of mineral hazard yet with different severities of interstitial lung abnormalities.

Fifty consecutive patients from Yunnan Cancer Hospital diagnosed with lung cancer who 1) underwent lobectomy between 2015 and 2019; 2) had a chest high-resolution computed tomography (HRCT) scan within 3 months before surgery; and 3) were not under chemotherapy or radiotherapy before surgery, were selected as cases. Surgical resections of neoplastic tissue along with para-carcinoma tissue and peribronchial lymph node tissue were obtained from lung cancer cases. Meanwhile, 30 patients diagnosed with inflammatory pseudotumor who underwent curative resection for lung and had a chest HRCT scan within 3 months before surgery were selected as controls, of which the surgical resections of normal respiratory epithelium tissue and peribronchial lymph node tissue were obtained. Besides chest HRCT scanning images and the corresponding pathological diagnoses, all other data of clinical

characteristics (including age at the time of surgery, marital status, occupational history, smoking history, alcohol drinking history, chronic respiratory disease conditions, and family history of lung cancer) were also retrieved from Yunnan Cancer Hospital. Written informed consent was obtained from all participants, and ethical approval was granted by the hospital's Institutional Review Board (IRB approval no.: KYCS2021113).

2.2. ILA severity grading by chest HRCT

ILAs were defined as radiologic abnormalities of lung nodules found incidentally on chest HRCT potentially related to interstitial lung disease (Chae et al., 2021). Two thoracic radiologists (GL and YC) conducted a retrospective analysis of preoperative chest HRCT images in a blinded manner. The following abnormality features were evaluated based on the Fleischner Society Nomenclature recommendations and similar studies (Hansell et al., 2008): distribution of nodules, interlobular septal thickening, consolidation, ground-glass nodules (GGNs), crazy-paving pattern, and reverse halo sign. To quantify the extent of pulmonary abnormalities in HRCT images, we employed a severity score developed by Elmokadem Ali et al.: 0 for normal lung; 1 for < 25 % lung abnormalities; 2 for 25–49 % abnormalities; and 3 for ≥ 50 % lung abnormalities (Elmokadem et al., 2022).

2.3. Histopathological grading of anthracotic pigments in lung tissues

Chest HRCT findings were complemented by histopathological diagnoses following lung surgery. In the context of chronic exposure, anthracotic pigment may co-occur with interstitial fibrosis, inflammation, and dust-related ILAs, and thus may serve as a surrogate marker of particulate exposure that predisposes to ILAs (Zhai and Yu, 2022). Two certified pathologists (ZH and JR), who were blinded to the chest HRCT results and patients' medical information, examined the extent of anthracotic pigments in each slide of lung resection specimens (see detailed procedures of preparing pathological sections in [Supplementary Methods](#)) under the light microscope (LM). A previously published histopathologic scoring system for coal workers' pneumoconiosis (CWP) to assess the size and extent of anthracotic pigments in the lung was adopted; a scale of 0–3 represented increasing severity of dust-related ILAs: 0 (absent), 1 (mild), 2 (moderate), and 3 (high) (Cohen et al., 2016; Vallyathan et al., 2011). Any discrepancies in HRCT scoring between the two pathologists were resolved through consensus discussion.

2.4. Characterization of birefringent mineral deposits in lung tissues

Polarized light microscope (PLM) was employed to examine birefringent mineral deposits in the lung specimens. A scoring system developed by Green et al. was leveraged to quantify the extent and distribution of mineral particles within the tissue samples at 100x magnification using the following criterion: 0 (absent), 1 (occasional), 2 (moderate), and 3 (high) (Green et al., 1997). Transmission electron microscope (TEM) was applied to observe and analyze the morphology of the mineral deposits in the lung tissues. Surgery tissue blocks of 1 mm in length were collected and fixed in glutaraldehyde in cacodylate buffer, postfixed and stained by uranyl acetate and osmium tetroxide. Then the pre-processed samples were further cut into 100 nm thick ultra-thin sections, stained with lead citrate, and examined under TEM-EDS (see detailed procedures of sample preparation for TEM analysis in [Supplementary Methods](#)).

2.5. Statistical analysis

Continuous variables were expressed as mean $\pm$ standard deviation (SD) and categorical variables were expressed as counts or percentages. Differences in demographic features, ILA severity and mineral deposits between lung cancer cases and pseudotumor controls were examined by

t test or Chi-square test, where appropriate. Statistical significance was assessed using a two-tailed test with an alpha level of 0.05.

3. Results

3.1. Demographic features of cases and controls

The average ages of lung cancer patients and pseudotumor patients were 49.8 ± 5.3 and 48.0 ± 9.8 years, respectively. All the 50 lung cancer cases were diagnosed as lung adenocarcinoma. For pseudotumor patients, 14 were diagnosed with granulomatous inflammation, 7 with inflammatory myofibroblastic tumor (IMT), and 9 with chronic, unspecified inflammation. Table 1 displays the demographic characteristics of our study participants. Overall, the lung cancer group (cases) and pseudotumor group (controls) were comparable in age, body mass index (BMI), smoking and alcohol drinking status, family history of lung cancer, and other comorbid diseases. None of the lung cancer patients nor pseudotumor patients have a history of tobacco smoking, alcohol drinking, industrial exposure history, or pulmonary tuberculosis.

3.2. ILAs in chest HRCT scans

In HRCT evaluation, the predominant ILA features identified in both lung cancer and pseudotumor groups included GGNs, linear opacities, consolidation, interlobular septal thickening, and the crazy-paving pattern. Semi-quantitative assessment revealed that lung cancer patients demonstrated a significantly greater extent of ILAs compared to pseudotumor controls (Table 2; $p < 0.05$). Specifically, lung cancer cases were characterized by broader areas of GGN involvement, more prominent and widespread interlobular septal thickening, and a higher incidence of crazy-paving appearance, suggesting more advanced lesions as presented in Fig. 1.

3.3. Anthracotic pigments and mineral deposits in lung tissues

In LM evaluation of lung tissue specimens, lung cancer patients demonstrated a significantly greater extent of anthracotic pigments compared to pseudotumor controls in both lung parenchyma and peribronchial lymph nodes (Table 3; $p < 0.05$). Under PLM, peribronchial lymph nodes of lung cancer patients demonstrated a significantly greater extent of birefringent mineral deposits than those of pseudotumor controls (Table 4; $p < 0.05$).

Histopathological images of carcinoma, para-carcinoma, and peribronchial lymph node tissues, respectively, from one lung cancer patient under LM and PLM are presented in Fig. 2. Fibroplasia was particularly

Table 1

Demographic comparison of lung cancer group (cases) and pseudotumor group (controls) among non-smoking female patients from Xuan Wei, China¹.

Characteristics	Lung Cancer Group (n = 50)	Pseudotumor Group (n = 30)	<i>p</i> -value ²
Age (years)	49.8 (5.3)	48.0 (9.8)	0.29
BMI (kg/m ²)	23.2 (3.3)	23.6 (2.7)	0.60
Married	50 (100.0 %)	30 (100.0 %)	
Alcohol drinking history	0 (0 %)	0 (0 %)	
Family lung cancer history	11 (22.0 %)	3 (10.0 %)	0.17
Other comorbid diseases			
Pulmonary tuberculosis	0 (0 %)	0 (0 %)	
COPD	4 (8.0 %)	5 (16.7 %)	0.23
Chronic hypertension	2 (4.0 %)	2 (6.7 %)	0.60

Abbreviations: BMI, body mass index; COPD, Chronic Obstructive Pulmonary Disease.

¹ All the female participants were confirmed to be homemakers with no history of other occupational exposures.

² The *p*-value was generated by *t* test or Chi-square test, where appropriate.

Table 2

Comparison of ILA severity through chest HRCT between lung cancer cases and pseudotumor controls among non-smoking female patients from Xuan Wei, China.

ILA Severity Evaluated by Chest HRCT ¹	Lung Cancer Group (n = 50)	Pseudotumor Group (n = 30)
0	0 (0 %)	0 (0 %)
1	38 (76.0 %)	28 (93.3 %)
2	9 (18.0 %)	1 (3.3 %)
3	3 (6.0 %)	1 (3.3 %)
<i>p</i> -value ²	0.04	

Abbreviations: ILA, interstitial lung abnormalities; HRCT, high-resolution computed tomography

¹ Scores of 0–3 represent normal lung, < 25 % lung abnormalities, 25–49 % abnormalities, and ≥ 50 abnormalities, respectively.

² The *p*-value was derived from Chi-square test.

extensive in para-carcinoma and peribronchial lymph node tissues. Birefringent mineral deposits were observed in regions with anthracotic pigments within all three types of tissue samples (especially peribronchial lymph nodes). They were distributed as focal clusters or widely scattered single particles. Consistent with the observations under LM and PLM, extensive collagen fibers and fibrotic remodeling were more readily visible in para-carcinoma and peribronchial lymph nodes than in cancerous tissues under TEM (Fig. 3). Mineral deposition in peribronchial lymph nodes was often massive. The occasionally spotted mineral fibers in cancerous tissues were often wavy, which could be residues of acicular berthierine-chamosite; in contrast, some mineral fibers in para-carcinoma and peribronchial lymph node tissues still appeared acicular, resembling the morphology of berthierine-chamosite in Late Permian coal (Kogure T et al., 2019; Li et al., 2021).

4. Discussion

In this study, we found that ILAs and mineral deposits in lungs are common in both lung cancer cases and pseudotumor cases from Xuan Wei, China. Lung cancer cases tend to have more severe fibrotic changes and heavier mineral deposits compared with their pseudotumor controls. The mineral deposits, especially in the peribronchial lymph nodes, are morphologically resembling the acicular berthierine-chamosite in Late Permian coal, the uppermost seam used in high-risk regions (Dai et al., 2008; Kogure et al., 2019; Li et al., 2021). Berthierine belongs to the serpentine mineral group, as does chrysotile asbestos; berthierine is Fe-rich whereas chrysotile is Mg-rich.

Previous studies have demonstrated that the presence of ILAs is associated with an increase in lung cancer complications and mortality risk (Im et al., 2023). Studies on idiopathic pulmonary fibrosis have also suggested that lung scarring may serve as a risk factor for lung carcinogenesis (Karampitsakos et al., 2017). In our study, while the extent and severity of fibrotic changes differ between lung cancer cases and pseudotumor controls from the same endemic region where coal smoke exposure is prevalent, both groups share some similar characteristics of mineral deposition in lungs. Future longitudinal follow-up studies are warranted to determine if pseudotumor patients with mild ILAs represent early signs of this mineral-fibrosis pathway and are at high risk for progressing to lung cancer. Our histopathological observation revealed close linkage between fibrotic foci and mineral deposition in the para-carcinoma lung tissues. Moreover, birefringent mineral deposits and ILAs were more prominent in the peribronchial lymph nodes of lung cancer patients than in pseudotumor patients. It is likely that impaired lymph node defenses could exacerbate the mineral carcinogen deposits in the parenchyma and contribute further to cancer development. Fibrosis of the peribronchial lymph nodes probably preceded lung fibrosis in 4384 German Uranium miners with lung cancer (Mielke et al., 2018; Taeger et al., 2011). Similarly, non-smoking housewives of Xuan Wei, China with habitual exposure to indoor air pollution from smoky

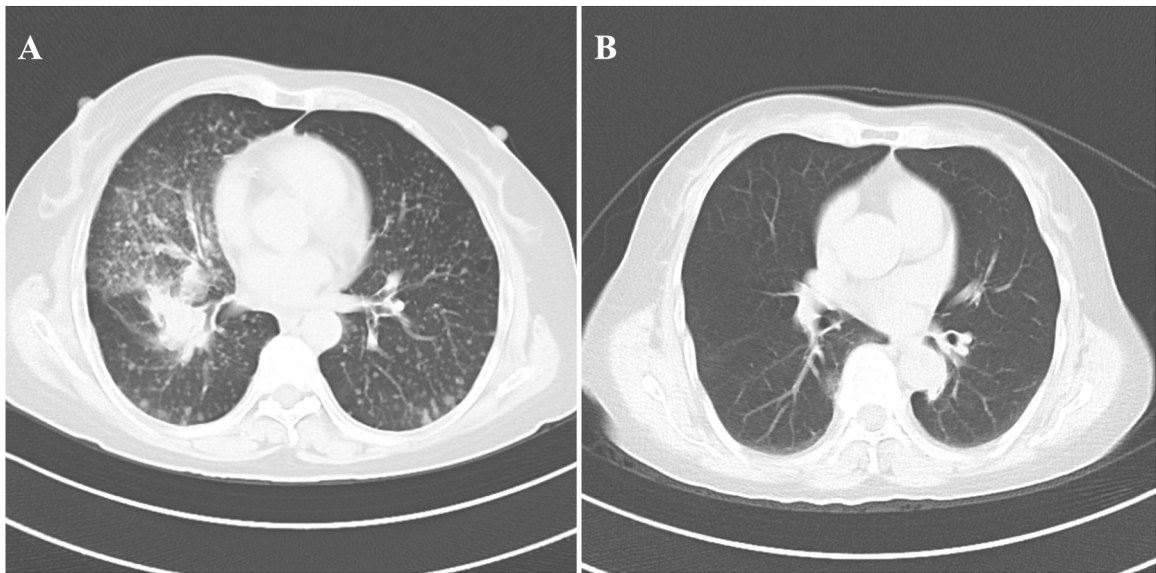


Fig. 1. HRCT evaluation of interstitial lung abnormalities in lung cancer case and pseudotumor control among non-smoking female patients from Xuan Wei, China. A) Lung cancer patients were characterized by broader areas of ground-glass nodules, more prominent interlobular septal thickening, and a higher degree of crazy-paving patterns, in comparison with B) pseudotumor patients.

Table 3
Comparison of ILA severity through LM observation of lung tissue specimens between lung cancer cases and pseudotumor controls among non-smoking female patients from Xuan Wei, China.

Anthracotic pigments Evaluated by LM ¹	Lung Parenchyma		Peribronchial Lymph Nodes ²	
	Lung cancer (n = 50)	Pseudotumor (n = 30)	Lung cancer (n = 46)	Pseudotumor (n = 8)
0	4 (8.0 %)	6 (20.0 %)	0 (0.0 %)	0 (0.0 %)
1	31 (62.0 %)	20 (66.7 %)	0 (0.0 %)	1 (12.5 %)
2	12 (24.0 %)	4 (13.3 %)	3 (6.5 %)	0 (0.0 %)
3	3 (6.0 %)	0 (0.0 %)	43 (93.5 %)	7 (87.5 %)
p-value ³	0.02		0.02	

Abbreviations: ILA, interstitial lung abnormalities; LM, light microscope
¹ Scores of 0–3 represent increasing extent of anthracotic pigments: 0 for absent, 1 for mild, 2 for moderate, and 3 for high.
² Tissue samples of peribronchial lymph nodes were not available from all the participants.
³ The p-value was generated by Chi-square test.

coal combustion could have suffered from repeated lung injury, chronic inflammation, tissue remodeling (scarring), and impaired lymph node defenses, which creates a microenvironment of malignant transformation (Elias et al., 1990).

The birefringence of mineral deposits in the lungs of this lung cancer epidemic area indicates etiological causes similar to those in occupational settings with crystalline silica (quartz) and silicate mineral exposures. No silicotic nodules, however, were found in either the lung cancer or the pseudotumor patients in this study. Fine quartz deposits are occasionally visible in the lung tissues, but the extent is much smaller than the deposits of fibrous silicate minerals that morphologically resemble berthierine-chamosite, a major mineral identified in the Late Permian coal of Xuan Wei, China (Kogure et al., 2019; Li et al., 2021). Accurate identification of such minerals typically requires golden standard techniques including X-Ray Diffraction (XRD) and Selected Area Electron Diffraction (SAED). However, this task is challenging in human lung tissues because the fibers have largely lost their crystallinity and

Table 4
Comparison of birefringent mineral deposits through PLM between lung cancer cases and pseudotumor controls among non-smoking female patients from Xuan Wei, China.

Mineral Deposits Evaluated by PLM ¹	Lung Parenchyma		Peribronchial Lymph Nodes ²	
	Lung cancer (n = 50)	Pseudotumor (n = 30)	Lung cancer (n = 45)	Pseudotumor (n = 14)
0	26 (52.0 %)	18 (60.0 %)	0 (0.0 %)	7 (50.0 %)
1	20 (40.0 %)	10 (33.3 %)	16 (35.6 %)	4 (28.6 %)
2	4 (8.0 %)	2 (6.7 %)	15 (33.3 %)	2 (14.3 %)
3	0 (0.0 %)	0 (0.0 %)	14 (31.1 %)	1 (7.1 %)
p-value ³	0.30		0.02	

Abbreviations: PLM, polarized light microscope
¹ Scores of 0–3 represent increasing levels of birefringent mineral deposits within the tissue samples at 100x magnification: 0 for absent, 1 for occasional, 2 for moderate, and 3 for high.
² Tissue samples of peribronchial lymph nodes were not available from all the participants.
³ The p-value was generated by Chi-square test.

become amorphous, undergoing thermal transformation during coal combustion followed by acidic degradation within macrophage autolysosomes. Our TEM results showed that while mineral fibers in the peribronchial lymph nodes still appear acicular, those found in the cancerous tissues were often wavy. The different patterns of mineral deposits in cancerous tissues and peribronchial lymph nodes probably offer a histologic snapshot of the prolonged pathogenic process, capturing both the body's attempt to clear inhaled particles and the ultimate failure leading to malignancy. Our inference on this suspected fibrous mineral and its associated underlying carcinogenic pathway must be corroborated by evidence from prospective and interventional studies.

Both the Fe²⁺-rich berthierine and chamosite were found to be the most active among all iron ores in producing free radicals within a few hours in the cell-free phosphate buffer medium. In contrast, quartz and

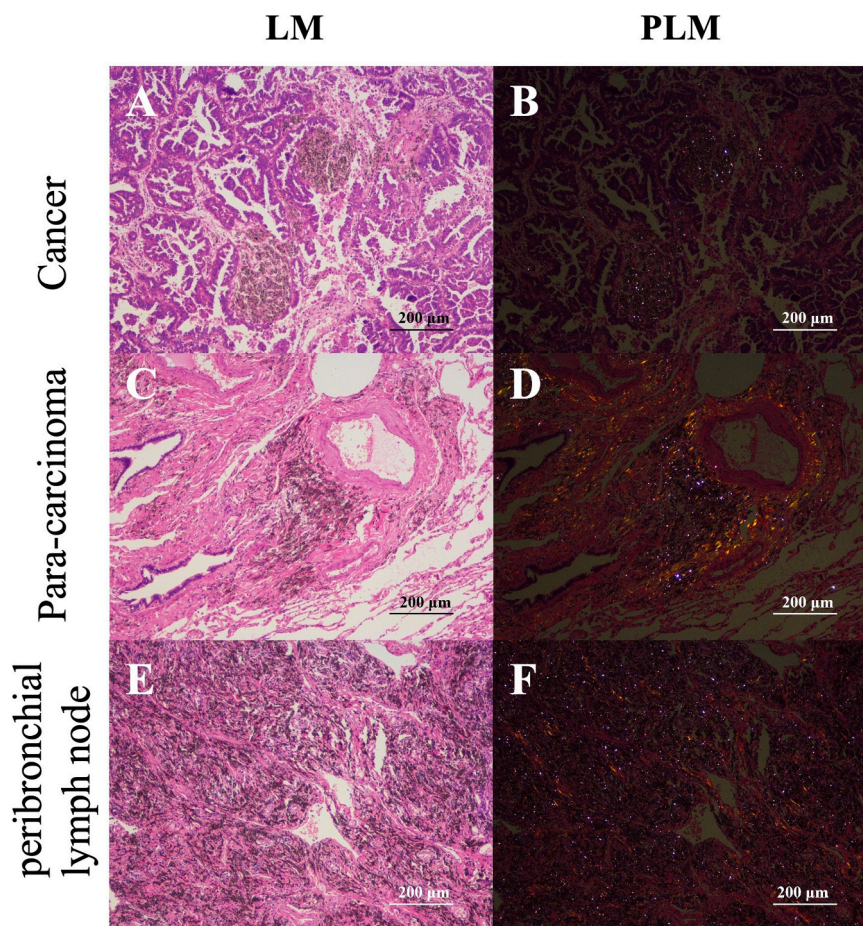


Fig. 2. Histopathological examination of anthracotic pigments by LM and birefringent mineral deposits by PLM in the lung tissues of a non-smoking female lung adenocarcinoma patient from Xuan Wei, China. A), C), and E) show anthracotic pigments (dark black colored fine granular particles) in cancer tissues, para-carcinoma tissues, and lymph node tissues, respectively. B), D), and F) show the corresponding view of mineral deposits in the same field as A), C), and E), respectively, under PLM. Note that anthracotic pigments are surrounded with extensive fibroplasia, especially in para-carcinoma tissues C) and D).

phyllosilicates poor in Fe^{2+} were weakly or non-active (Costa et al., 1989). Divalent iron, an electron donor, on the asbestos fibers has been implicated in the carcinogenesis process (Weinberg, 1999). This lays the foundation for understanding the carcinogenic potential of asbestos-like berthierine-chamosite derived from local coal. But the housewives in our study setting are exposed to coal smoke, not coal dust or any intact berthierine-chamosite minerals. It is not known whether the intact minerals and the modified ones by heat and acids would carry equivalent carcinogenic potencies. Future controlled animal experiments, which are on-going work in our laboratory, would help address this research question building upon the human epidemiology and pathology evidence presented here. Our findings from this lung cancer epidemic of Xuan Wei, China could generate more mechanistic insights into the carcinogenic potency of chrysotile asbestos and other minerals that share similar characteristics with berthierine-chamosite in Late Permian coal, thereby advising targeted intervention strategies locally and globally.

The limitations of this study require due discussion. First, the use of pseudotumor patients as control group, while essential for obtaining surgically resected lung tissues for analysis, limits the generalizability of our findings. The comparison of two pathologies here, “disease vs. disease” relationship, was not directly on the etiology of either pathology which has to be examined by community-based studies on “disease vs. non-disease” relationship, thereby limiting causal inference. Future studies could combine community-based controls with autopsy lung tissue samples of non-respiratory deaths to isolate the specific effects of environmental exposure. Second, the characterization of silicate

minerals is based on microscopy techniques (PLM and TEM). Although the morphology of particles closely resembles berthierine-chamosite from local smoky coal, the absence of confirmatory crystallographic data (e.g., from XRD or SAED) necessitates cautious interpretation. Third, the sample size is relatively small, particularly for the subgroup analyses of peribronchial lymph nodes due to limited surgical availability. This may reduce the statistical power to measure differences and pose challenges in extrapolating our findings; yet, the qualitative patterns observed still provide a valuable foundation for future larger-scale studies. Fourth, detailed historical exposure data at the individual level are lacking; however, our analysis of silicate particle burden in lung tissue samples represents a biologically relevant measure of long-term cumulative exposure. More importantly, our research team is building an Indoor Coal User Cohort (ICUC) to systematically document the coal use history and longitudinally track lung pathology and cancer development. Finally, this study is of retrospective observational nature and therefore susceptible to residue confounding. By including only non-smoking females who were lifelong housewives as study participants, we controlled for the major confounders of smoking, gender, and occupational history. Other key factors such as age, family history of lung cancer, and comorbid COPD were found balanced between cases and controls.

5. Conclusion

Interstitial lung abnormalities (ILAs) and some fibrous mineral deposits are common in both lung cancer and pseudotumor patients from

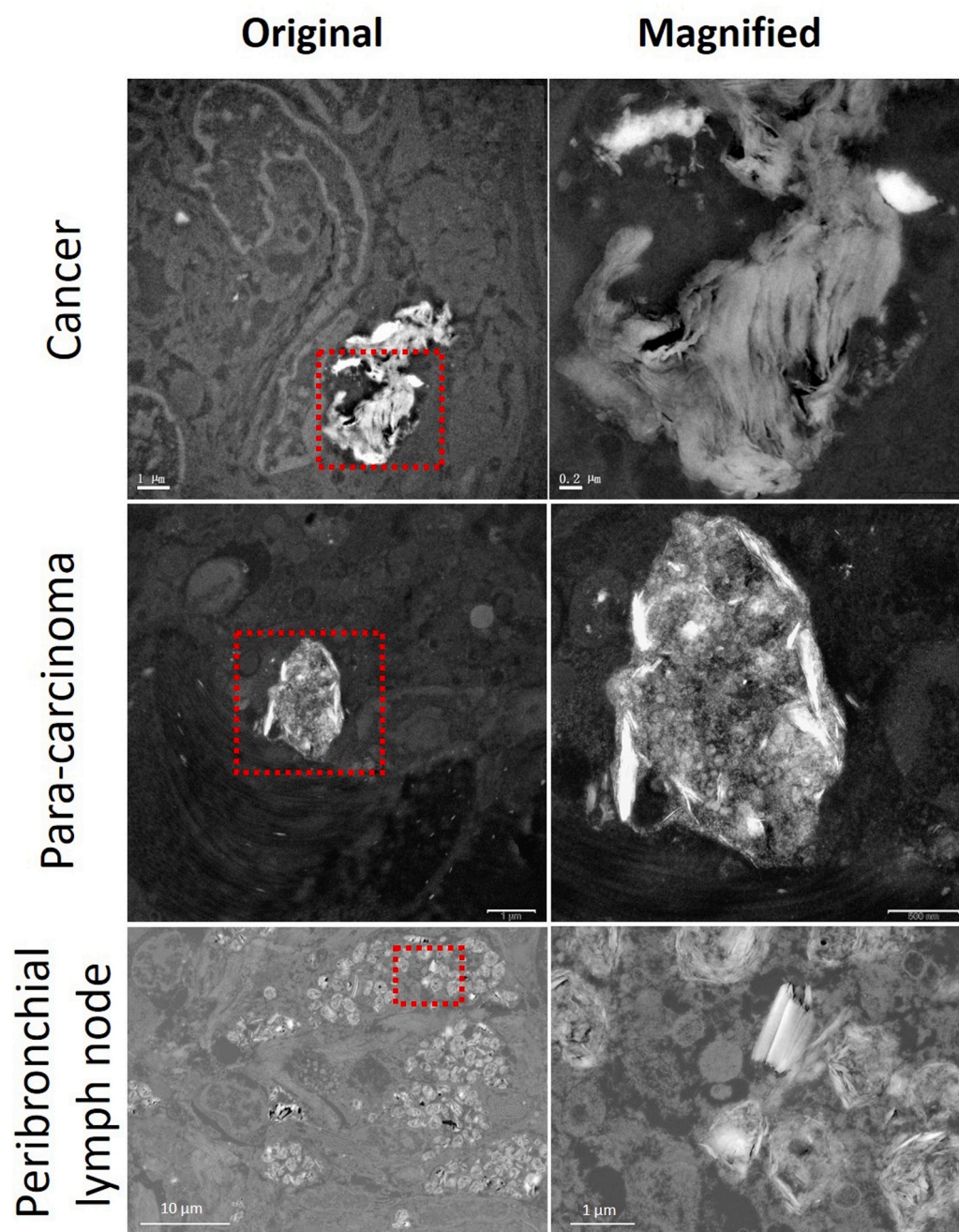


Fig. 3. Inverted TEM micrographs to highlight the mineral deposits in cancerous tissue, para-carcinoma tissue, and peribronchial lymph nodes, respectively, of non-smoking female lung cancer patients from Xuan Wei, China. Fibrous mineral deposition and fibrotic remodeling in peribronchial lymph nodes are more extensive than in cancer or para-carcinoma tissues. While the mineral fibers in cancerous tissues are often wavy, those in para-carcinoma and peribronchial lymph nodes still appear acicular and resemble the morphology of berthierine-chamosite in Late Permian coal (Kogure T et al., 2019; Li et al., 2021).

Xuan Wei, China, while it tends to be more severe in the former group. The peribronchial lymph node defense probably has been severely impaired in the lung cancer cases in comparison with their pseudotumor controls. The fibrous mineral in lungs morphologically resemble inter-layered berthierine-chamosite identified as a major mineral in Late Permian coal of Xuan Wei, China. As the Fe^{2+} -rich berthierine is in the same serpentine group with Mg^{2+} -rich chrysotile asbestos, etiology findings in this lung cancer epidemic would bear great implications in both local and global contexts.

CRediT authorship contribution statement

Zheshen Han: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Software, Visualization. **Zaoxiu Hu:** Validation, Methodology, Investigation, Data curation. **Zhouxi Fang:** Investigation, Formal analysis, Data curation, Conceptualization. **Ying Chen:** Investigation, Funding acquisition. **Guangjian Li:** Investigation, Formal analysis. **Jingli Ren:** Investigation, Formal analysis. **Jinhui Li:** Investigation, Methodology. **Hongli Lin:** Investigation. **Renyue Ji:** Investigation, Data curation. **Jian Zhang:** Investigation. **Weijun Li:** Investigation. **Fang Guo:** Writing – review & editing, Validation, Project administration, Methodology, Investigation. **Yunchao Huang:** Data

curation, Resources, Funding acquisition. **Linwei Tian:** Writing – review & editing, Conceptualization, Supervision, Validation, Resources, Project administration, Methodology, Funding acquisition.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.ecoenv.2025.119604](https://doi.org/10.1016/j.ecoenv.2025.119604).

Data availability

Data will be made available on request.

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